



Islamic University of Technology

Department of Computer Science and Engineering (CSE)

CSE 4206: Digital Logic Design Lab
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Faisal Hussain, Assistant Professor, CSE
Aashnan Rahman, Jr. Lecturer, CSE

Lab 5

Combinational Circuits II

Contents

1	Introduction	2
2	BCD to 7 Segment Display	2
2.1	Objectives	3
3	Tasks	4
3.1	Design a 4-bit Adder-Subtractor Circuit using 7 segment display	4
3.1.1	Instructions	4
3.2	A Very Mid Combinational Circuit	5
3.2.1	Instructions	5
4	Instructions:	6

1 Introduction

In the previous lab session, we focused on the simulation of basic arithmetic circuits, specifically adders and subtractors. These circuits form the foundation of many digital systems. In today's lab, we will build upon that knowledge by simulating more complex combinational circuits. In this lab we will practice designing a few combinational circuits.

Warning: If you are caught copying another classmate's lab task, you will receive a zero for that specific lab.

2 BCD to 7 Segment Display

In today's lab, you will also need to display BCD (Binary-Coded Decimal) values on a 7-segment display. For this, you may use the 4511 BCD to 7-Segment IC.

The IC takes a 4-bit BCD input on pins A (pin 7), B (pin 1), C (pin 2), and D (pin 6):

- A is the Least Significant Bit (LSB)
- D is the Most Significant Bit (MSB)
- The outputs (pins 9 to 15) control the segments a to g of the 7-segment display.

Table 1: 4511 BCD to 7-Segment Display Driver – Key Pin Functions

Pin	Name	Set To	Description
1	B	Input	BCD Input Bit (Bit 1)
2	C	Input	BCD Input Bit (Bit 2)
3	LT	HIGH	Lamp Test – Keep HIGH to disable segment test mode
4	BI	HIGH	Blanking Input – Keep HIGH to enable output
5	LE	LOW	Latch Enable – Keep LOW for transparent input-to-output behavior
6	D	Input	BCD Input Bit (MSB – Bit 3)
7	A	Input	BCD Input Bit (LSB – Bit 0)
8	GND	GND	Ground connection (0V)
9–15	a–g	Output	Segment control outputs for 7-segment display
16	Vcc	+5V	Power supply

And for the seven segment display, make sure to ground (0) the 8th pin. Alternatively you can use something simpler like this:

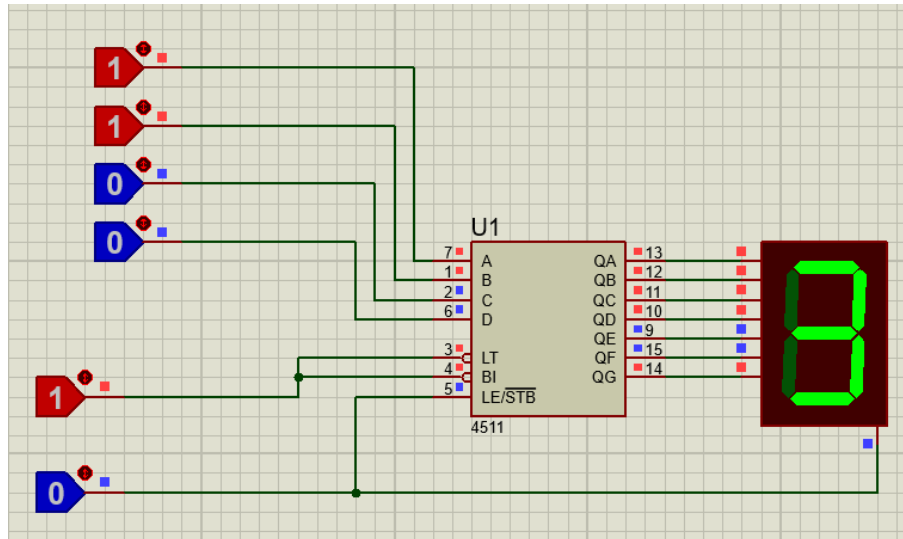


Figure 1: BCD to 7 segment display

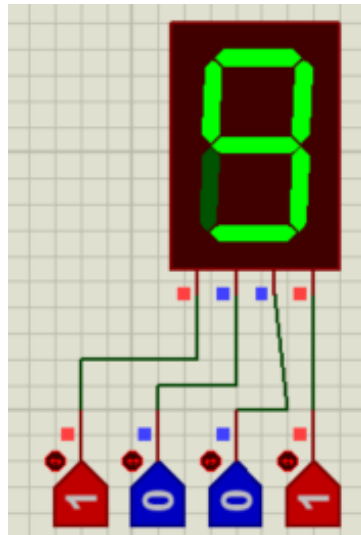


Figure 2: BCD to 7 segment display

2.1 Objectives

The objectives of this experiment are as follows:

- Perform 2-digit BCD addition with proper correction and display.
- Compare two BCD digits using basic logic gates.
- Use BCD to 7-segment decoder (e.g., 4511) to drive displays.

3 Tasks

Today's lab task are as follows -

3.1 Design a 4-bit Adder-Subtractor Circuit using 7 segment display

In this task, you will implement a **4-bit adder-subtractor circuit** and display both the **inputs and output using 7-segment displays**.

3.1.1 Instructions

1. Use two 4-bit binary numbers (**A** and **B**) as inputs, where each input represents a valid decimal digit (0–9).
2. Construct the adder-subtractor using 4-bit full adder blocks and a mode control input to select between addition and subtraction.
3. Implement the subtraction operation by complementing the **B** input and adding 1 using basic logic gates.
4. Ensure that the subtraction operation is performed only for cases where $\mathbf{A} \geq \mathbf{B}$, and the addition operation is performed only when $\mathbf{A} + \mathbf{B}$ does not exceed 9.
5. Use BCD to 7-segment decoder circuits to display both input values (**A** and **B**) and the output result on 7-segment displays.
6. Test the circuit with various valid input combinations to verify correct addition and subtraction operations.

Allowed Components

- 2-bit full adder blocks
- Basic logic gates (AND, OR, NOT, XOR)
- BCD to 7-segment decoder modules
- Input switches or binary input sources
- 7-segment display units

3.2 A Very Mid Combinational Circuit

In this task, you will implement a **4-bit binary subtractor circuit** using the **diminished radix complement method** and display the result using **7-segment displays**.

3.2.1 Instructions

1. Use two 4-bit binary numbers (**A** and **B**) as inputs, where **B** is always subtracted from **A**.
2. Design the subtraction circuit using a 4-bit adder block and necessary logic gates to perform subtraction using the diminished radix complement method.
3. The circuit must handle both cases where $A \geq B$ and $A \leq B$ by determining the **correct magnitude and sign of the result**.
4. The output should consist of a **4-bit value** representing the **magnitude of the difference** and **one additional** bit representing the **sign** of the result (positive or negative).
5. **Do not use separate output paths for positive and negative results; the sign bit must indicate the result sign.**
6. Use BCD to 7-segment decoder circuits to display the input values (**A** and **B**), and magnitude of the result, and use an LED to show the sign of the output.
7. Test the circuit with various 4-bit input combinations to verify correct subtraction and sign indication.
8. You can reuse parts of previous task.

Allowed Components

- Basic logic gates (AND, OR, NOT, XOR)
- BCD to 7-segment decoder (or manual segment control using gates)
- 7-segment display
- Input switches or binary sources

Invalid Solution

- **Two different answers for positive and negative results**

4 Instructions:

1. **File Naming Format:** ID_L05T01 and ID_L05T02.
2. For task 2 you have to come up with the explanation of how you are showing the result in a seven segment display.
3. You will get 1 hour to complete these tasks.
4. The rubrics for today's lab task can be found [here](#).